A Hierarchical Multi-Modal Framework for Continuous Sediment Grain-Size Estimation using Sentinel-2 Imagery

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Abstract. Accurate description of the size of the sediment grains is essential to study the coastal geomorphology, hydrodynamic energy regimes, and benthic habitat distribution. Although conventional semantic segmentation groups features of coastal areas into discrete classes (e.g., sand vs. mud), in many instances it is not able to represent the gradual grains of granular differences present in the natural intertidal systems. The present paper suggests a hierarchical machine learning architecture that combines deep Convolutional neural networks (CNNs) with regression ensembles in order to predict continuous percentages of grain-size. And we then use a DeepLabv3+ network with MobileNetV2 backbone to carefully extract the intertidal area of Chandipur coast using Sentinel-2 satellite data with a Mean Intersection over Union (MeanIoU) of 0.8860. Afterward, a Random Forest regressor that has been trained on spectral values of visible and Short-Wave Infrared (SWIR) bands is used to estimate the exact ratio of sand to mud in that pixel. We have a regression model with a Root Mean Square Error (RMSE) of 6.42 that is validated using in-situ geological samples. Our model illustrates that the fine-scale sedimentological differences are best reflected in regression-based modeling and not in categorical classification alone; hence, providing a high-resolution view of geological analysis.

Keywords: Remote Sensing · Deep Learning · Random Forest · Sediment Grain Size · DeepLabv3+ · Sentinel-2.

1 Introduction

It can be stated that intertidal flats are the most dynamic and complex coastal ecosystems in the world that act as barriers to storm surges, coastal pollutant filters, and exclusive habitats of various benthic communities as well [1]. Proper mapping of these areas is critical to environmental surveillance especially with the growing sea level and the growing anthropogenic stress. Nevertheless, the classification of intertidal sediments has a specific range of challenges to mapping in contrast to terrestrial land cover. The subtle differences in hydrodynamic energy control the distribution of the sediment, so that there are no sharp surfaces between the sand and mud, but smooth gradients of grain sizes.

The Chandipur coast in Odisha, India, is an extreme environment of such a kind. It has a low-gradient seabed with a macrotidal regime, in which the tide reaches 5 km in depth at low tide. In the area, the differentiation of the visual appearances of the sediments is complicated by differences in moisture; wet sand may have a spectral similarity to mud, and drying mudflats may resemble sand.

The conventional remote sensing techniques have mainly depended on discrete classes of classification of algorithms such as the Support Vector Machines (SVM) or the Maximum Likelihood Classifiers [3]. These approaches separate pixels into mutually exclusive classes (e.g. Sandy Beach, Tidal Flat). Although this discretization can be used in large-scale mapping, it causes a serious loss of information. It does not represent the zone between sediment bodies- zones which are usually the most ecological and geological important. Moreover, the high-resolution satellite imagery has a salt-and-pepper noise that is hard to handle by pixel-based approaches (4).

However, as of recent, the accuracy of semantic segmentation has greatly advanced due to the state-of-the-art work in Deep Learning, specifically Convolutional Neural Networks (CNNs) such as U-Net (2) and DeepLabv3+ (which incorporate spatial context). Nevertheless, these sophisticated models often produce discrete labels of classes. They inform us of the location of the tidal flat, but not of the characteristics it has (e.g. what percentage of the grain is fine), which are also physical.

To solve this weakness we suggest an approach called a Hierarchical Multi-Modal Framework that would bridge the semantic versus physical property and understanding divide. We will consider the problem in two steps:

1. **Isolation (Segmentation):**The DeepLabv3+ model is a state-of-the-art tool that is used to segment the intertidal zone and confounding features in the background, such as the deep water and high density vegetation.
2. **Estimation (Regression):**Fitting a Random Forest Regressor in the known tidal regions to forecast continuous grain-size data using spectral reflectance.

The hybrid approach utilizes the spatial context sensitivity of Deep Learning to provide an accurate boundary demarcation and the tabular prediction capability of Ensemble Learning to provide accurate physical parameter predictions.

2 Related Work

2.1 Remote Sensing of Intertidal Zones

Early intertidal mapping studies extensively depended on multispectral indices such as the Normalized Difference Water Index (NDWI) in distinguishing land and water. Doerffer and Murphy [1]3 applications Landsat TM data and Principal Component Analysis (PCA) to differentiate between the types of sediment, which formed the basis of the conceptual connection between spectral reflectance and grain size of sediment. Nevertheless, these linear models tended to fail with the non-linear nature of varying moisture content in the macrotidal regions of the world oceans by nature

2.2 Deep Learning in Coastal Mapping

The Deep Learning changed the way coastal mapping was done. Kim et al. applied U-Net to UAV images to classify surface sediment, which was better than SVMs (2). This was also applied by Gargiulo et al. to the Synthetic Aperture Radar (SAR) data, where deep learning was used to map fluvial morphological units, which is cited in 4. Recently, DeepLabv3+, equipped with an Atrous Spatial Pyramid Pooling (ASPP), has become an architecture of choice when dealing with the multi-scale nature of coastal features, since it is capable of extracting one and capturing the fine-grained texture (local context) and broad coastal morphology (global context) at the same time.

2.3 Grain Size Estimation

While classification is common, continuous estimation of grain size is less explored. Dietze [6] proposed end-member modeling algorithms to deconvolute grain-size distributions. Recent studies have attempted to link optical reflectance directly to grain size using regression techniques, but often suffer from noise when applied to full satellite scenes without prior masking. Our work builds upon this by introducing a hierarchical step: we strictly mask the region of interest using deep learning before attempting regression, thereby reducing noise and false positives from non-sediment pixels.

3 Study Area and Data

3.1 Study Area

The research area is the Chandipur coast (approx. 21°27'N, 87°02'E) in the Balasore district of Odisha, India. The area is a special macrotidal habitat that has a semi-diurnal tidal cycle that is over 4 meters. The very low gradient of the continental shelf reveals kilometers of intertidal flats at the time of low tide. The Budhabalanga River estuary affects the sedimentology to form a complicated combination of fine silt, clays, and sands.

3.2 Satellite Data Acquisition

We utilized Copernicus Sentinel-2 Level-2A (L2A) imagery. The L2A product provides Bottom-Of-Atmosphere (BOA) reflectance, having been atmospherically corrected to minimize scattering effects. A critical aspect of our data curation was **Tidal Synchronization**. Using historical tide charts [7], we filtered the Sentinel-2 archive to select only those images acquired within ±2 hours of peak low tide. This rigorous filtering resulted in a high-quality dataset of 15 cloud-free scenes from 2017, 2018, and 2020.

We utilized five specific spectral bands known for their sensitivity to sediment and moisture properties (Table 1).

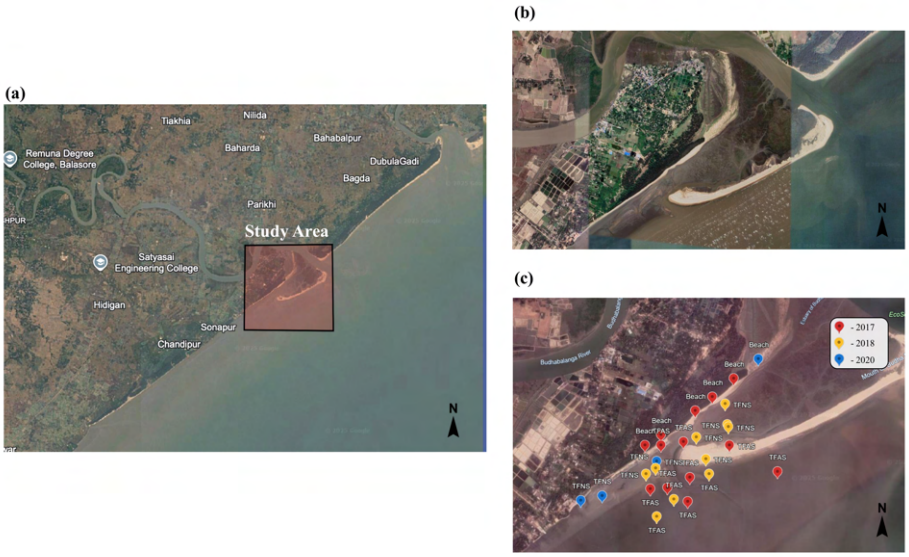

Fig. 1. Geographical setting of the Chandipur coastal study region. (a) Regional satellite imagery highlighting the boundaries of the study area. (b) Detailed close-up of the Chandipur coastline. (c) Spatial distribution of in-situ sediment sampling sites from the 2017, 2018, and 2020 campaigns.

Table 1. Sentinel-2 Spectral Bands used for Feature Extraction.

Band	Resolution	Central λ (nm)	Physical Relevance
B2 (Blue)	10m	490	Water penetration; sediment differentiation.
B3 (Green)	10m	560	Vegetation vigor; distinguishing algae.
B4 (Red)	10m	665	Soil brightness; absorption by vegetation.
B8 (NIR)	10m	842	Strong land-water contrast; moisture sensing.
B11 (SWIR)	20m	1610	Highly sensitive to soil moisture content.

3.3 In-Situ Ground Truth

In order to train and test our regression model, we used a dataset of sediment samples which were taken during field campaigns in 2017, 2018, and 2020. The analysis of the grain sizes in the laboratory gave the specific weight percentages of Sand and Mud of each location of the sample. This is our continuous data; it is our target variable of the regression stage (y). Table 2, Table 2 in appendix A, provides a representative sample of such geological information, showing the large range of sand-mud ratios of the various sub-environments.

Table 2. Results of In-Situ Grain Size Analysis (Sand and Mud Ratios) used for Regression Training.

Sub-environment	Year	Sand (%)	Mud (%)
TFNS (Near Shore)	2020	43.58	56.42
	2020	47.84	48.16
	2018	69.62	30.38
	2018	36.84	63.16
TFAS (Away Shore)	2018	64.68	35.32
	2017	94.90	5.10
	2017	88.54	11.55
	2017	91.01	8.99
Sandy Beach	2017	90.23	9.77
	2017	92.95	7.05
	2017	100.00	0.00
	2020	97.46	2.54

4 Methodology

The proposed framework operates on a sequential pipeline: (1) Deep Semantic Segmentation to isolate the sediment facies, and (2) Random Forest Regression to estimate grain size.

4.1 Stage 1: Deep Semantic Segmentation (Isolation)

This stage aims at developing a very precise binary mask that isolates the Potential Sediment and Water, Vegetation and Anthropogenic Features. The architecture we used is the DeepLabv3+.

DeepLabv3+ Architecture: DeepLabv3+ is an enhancement to standard U-Net architectures, whereby Atrous (Dilated) Convolutions are used. Normal convolution possesses a constant receptive field. Atrous convolution adds a rate parameter, r, which adds zeros in between the filter weights without making it more expensive to compute. In one dimension, where the signal to be processed by a filter is denoted by x[i], and the filter denoted by 1-D signal $x[i]$ and filter $w[k]$ of length K, the output $y[i]$ with rate r is:

$$y[i] = \sum_{k=1}^{K} x[i + r \cdot k]w[k] \tag{1}$$

This enables the network to obtain features at different scales. DeepLabv3+ uses this through the so-called **Atrous Spatial Pyramid Pooling (ASPP)** module, which uses parallel convolutions with rates $r = \{6, 12, 18\}$.

A backbone of the **MobileNetV2** that was pre-trained on ImageNet was used as our encoder. This is a light backbone that guarantees computational efficiency at the same time providing high feature extraction ability.

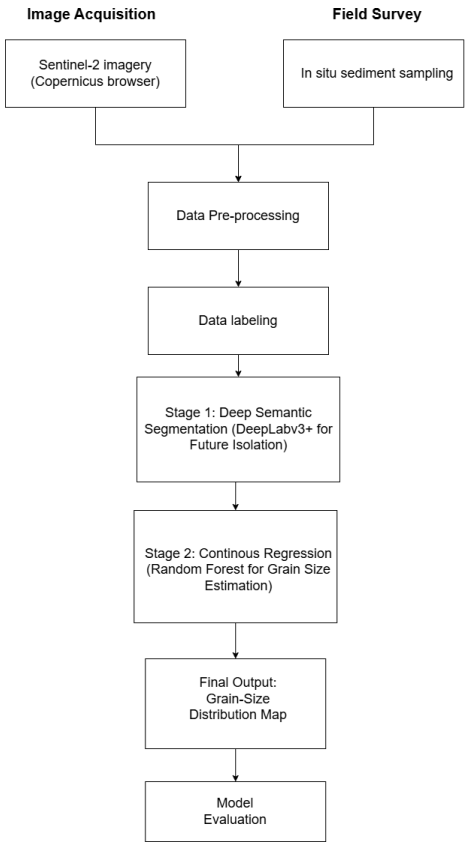

Fig. 2. The hierarchical workflow illustrating the multi-modal integration of Deep Learning and Ensemble Regression. The pipeline progresses from rigorous tidal synchronization and DeepLabv3+ segmentation (Stage 1) to Random Forest-based continuous grain-size estimation (Stage 2), validated against in-situ geological samples.

Training Configuration:

- **Loss Function:** To eliminate the issue of the huge disparity in classes with the largest class being the ocean (majority class) and the tidal flats, we adopted Focal Loss (L_{FL})and Dice Loss hybrid loss (FL) containing LFL and LDice.

$$L_{Total} = -\alpha(1-p)^{\gamma}\log(p) + \left(1 - \frac{2|A \cap B|}{|A| + |B|}\right) \tag{2}$$

  where $\gamma = 2.0$ focuses the model on hard-to-classify boundary pixels.
- **Optimizer:** Adam ($\eta = 0.0005$).
- **Epochs:** 30 epochs (leveraging transfer learning).

4.2 Stage 2: Continuous Grain-Size Estimation (Regression)

Pixels classified as *Tidal Flat Near Shore (TFNS)*, *Tidal Flat Away Shore (TFAS)*, or *Sandy Beach* by Stage 1 are passed to Stage 2. Here, we model the relationship between spectral reflectance and physical grain size.

Feature Extraction: For every valid sediment pixel p_{ij}, we extract a feature vector $X_{ij} = [B2, B3, B4, B8, B11]$. The target variable y_{ij} is the Sand Percentage (0-100%).

Random Forest Regressor: We utilized a Random Forest (RF) regressor. RF is an ensemble method based on Bootstrap Aggregation (Bagging). It constructs N decision trees during training and outputs the mean prediction of the individual trees.

$$\hat{y} = \frac{1}{N} \sum_{n=1}^{N} T_n(X) \tag{3}$$

where T_n is the n-th regression tree. Each split in the tree is determined by minimizing the Mean Squared Error (MSE) impurity:

$$\text{MSE} = \frac{1}{N_{node}} \sum_{i \in D_{node}} (y_i - \bar{y})^2 \tag{4}$$

We chose Random Forest over linear regression or simple neural networks because:

1. It is robust to outliers and noise in spectral data.
2. It handles non-linear relationships naturally.
3. It provides intrinsic "Feature Importance" metrics, helping us understand which bands drive the prediction.

5 Experimental Results

5.1 Stage 1: Segmentation Performance

DeepLabv3+ was tested on a set of held-out test sample (15% of the data). We contrasted it with a baseline U-Net model that was trained.

Table 3. Overall Performance Metrics on Test Set.

Metric	**U-Net**	**DeepLabv3+**
Overall Accuracy	94.84%	**97.76%**
Mean IoU	0.7927	**0.8860**
Cohen's Kappa	0.8828	**0.9518**

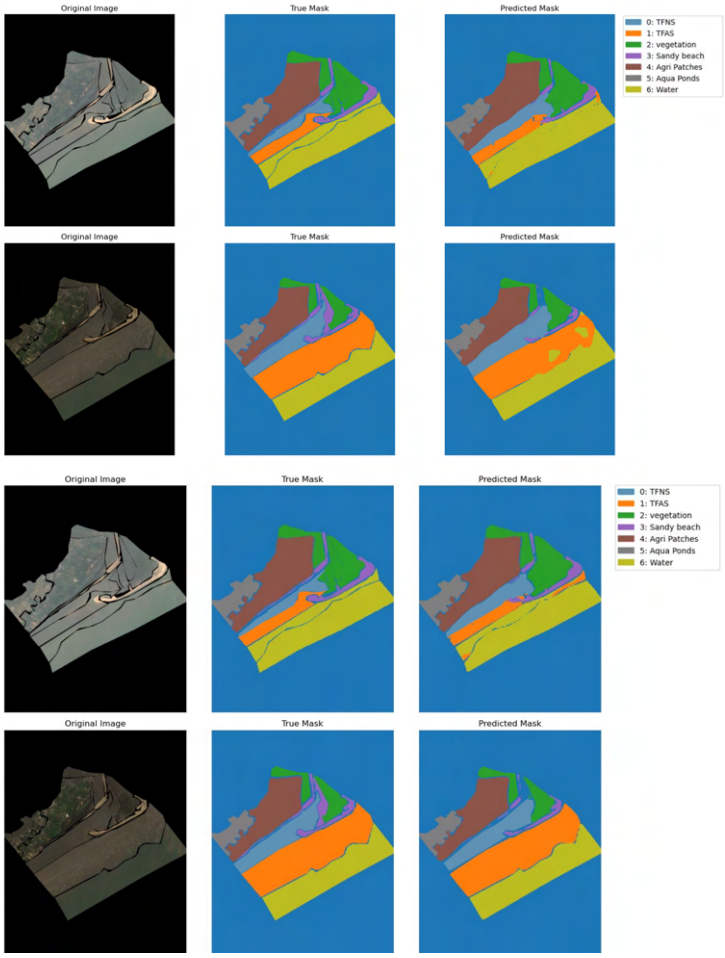

Fig. 3. Qualitative comparison of model predictions. (Left) U-Net output. (Right) DeepLabv3+ output. DeepLabv3+ shows improved edge definition and higher fidelity along complex contours.

In Table 3, DeepLabv3+ was much better than U-Net. The value of Kappa is high and it shows a near match with the ground truth. In order to learn about the areas of improvement, we studied the performance per class ((Table 4).

Most importantly, DeepLabv3+ improved tremendously in the class of "TFAS" and "Sandy Beach." These are the most spectrally related classes, and U-Net had issues distinguishing between the wet beach sand and tidal mud, but DeepLabv3+ context-aware ASPP module was able to distinguish between them depending on the area position against the shore.

Table 4. Per-Class F1-Score Comparison.

Class	Support	U-Net F1	DeepLabv3+ F1
TFNS (Mud)	648,264	0.98	**0.99**
TFAS (Tidal Flat)	46,253	0.75	**0.92**
Sandy Beach	13,964	0.62	**0.72**
Vegetation	30,849	0.81	**0.93**
Water	86,465	0.93	**0.97**

5.2 Stage 2: Regression Performance

The spectral features of pixels at the site of the in-situ samples were trained on the Random Forest regressor. Because of these small size of physical samples, we used Leave-One-Out Cross-Validation (LOOCV) to provide statistical soundness.

Accuracy Metrics: The regression equation had Coefficient of Determination (R^2) of **0.87** of the grain size variance explained by the satellite spectral characteristics. Root Mean Square Error was determined as:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_{obs} - y_{pred})^2} = 6.42\% \tag{5}$$

This implies that our model can estimate the sand percentage of any given pixel with an average error margin of roughly ±6.4%. Given that natural variability in sediment samples can be high, this is a highly promising result for remote monitoring.

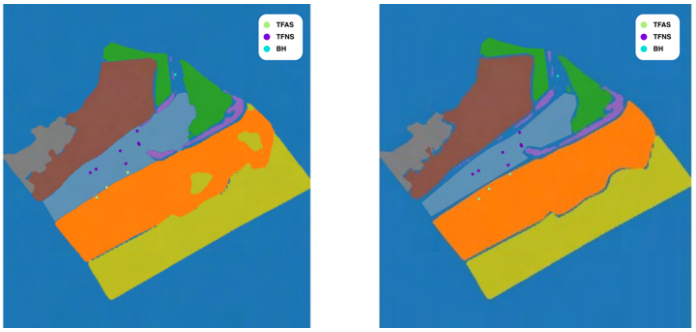

Fig. 4. Ground-truth validation using in-situ sediment points. (Left) U-Net classification. (Right) DeepLabv3+ classification. The overlay demonstrates that DeepLabv3+ provides a clearer distinction between TFNS and TFAS zones.

Feature Importance: The analysis by the Random Forest showed that the strongest predictor was the **SWIR Band (B11)** which contributed 40% to the decision importance. This is in line with the theory of sedimentology: fined grained muds contain more water (pore water) in comparison to coarse sands. B11 is a good proxy of moisture, and hence, grain size because water has very high absorption of SWIR light. The second most significant was the Red Band (B4) which was probably sensitive to the brightness difference between the dry silica sand and darker clay minerals.

6 Discussion

6.1 Hierarchical Efficacy

The findings confirm the bottom-up method. As a form of masking, we separated noise that would otherwise have confounded the regression model by using DeepLabv3+ on water and vegetation. Incorporation of the raw image would have made direct regression give nonsensical values of the grain size of the water pixels. The segmentation step also makes the regression semantically valid.

6.2 Error Analysis

Although the general RMSE was low, the model had increased error levels in the "Transition Zones" the slim strips where the sandy beach becomes the tidal flat. In these regions, Sentinel-2 (10m) pixel size can include a combination of sand and mud (mixed pixels), forming a spectral average which the model cannot understand. Moreover, the changes in surface water films just after a high tide may temporarily change reflectance, which proves the absolute need to ensure a strict tidal synchronization.

6.3 Implications for Coastal Management

With this framework, it is possible to generate continuous grain-size maps, which are much more useful than categorical maps in hydrodynamic modeling. These estimated grain sizes can help coastal engineers to estimate better the rate of sediment transport, erosion potential and appropriateness of the coastal infrastructure.

7 Conclusion and Future Work

This paper introduced a powerful, multi-modal model of mapping inter tidal sediment grain size. Combinations of the segmentation power of DeepLabv3+ with the estimation power of Random Forest allowed us to produce continuous maps of the grain-sized at the Chandipur coast with an accuracy of $R^2 = 0.87$. The research establishes that Sentinel-2 imagery by means of rigorous synchronization

with tidal data has enough spectral data to deduce the physical sediment characteristics. Further research will involve incorporation of the Synthetic Aperture Radar (SAR) data to enhance prediction accuracy in cloudy weather and the application of the UAV data to validate sub-meters.

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